LESSON 7: UNBOUNDED OPERATORS ON A HILBERT SPACE

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Self-adjoint unbounded operators are critical in quantum mechanics and functional analysis due to their unique properties. They represent physical observables, such as position and momentum, ensuring that measurement outcomes are real. As we will see self-adjoint unbounded operators ensure unitary evolution in quantum mechanics, preserving probabilities in quantum states.

1. Closed and Closable operators on a Hilbert space

In this section, we will recall the definitions of closed and closable operators on a Hilbert space. We will not provide the proofs of the main results stated in this section, but we will consider operators from Hilbert space H_1 to H_2.

Definition 1.1 (Definition of Unbounded Linear Operator)**.** *An unbounded linear operator from Hilbert spaces H_1 to H_2 is a pair $(A, D(A))$ where*

- *$D(A) \subset H_1$ is a linear subspace,*
- *$A : D(A) \to H_2$ is a linear map, i.e.,*
  (1) ***Additivity**: For all $x, y \in D(A)$,*
  $$A(x + y) = A(x) + A(y).$$
  (2) ***Homogeneity**: For all $x \in D(A)$ and all scalars $\alpha \in \mathbb{F}$ (where $\mathbb{F}$ is typically $\mathbb{R}$ or $\mathbb{C}$),*
  $$A(\alpha x) = \alpha A(x).$$

Definition 1.2. *A linear operator is densely defined if $D(A)$ is dense in H_1.*

Remark 1.3. *We will say "linear operator" (or simply "operator") for an unbounded linear operator when we specify the domain, since in this case no confusion will arise.*

Definition 1.4 (Kernel of a Linear Operator)**.** *Let $A : D(A) \subset H_1 \to H_2$ be a linear operator from $D(A)$ to a vector space Y. The kernel (or null space) of A, denoted by $\ker(A)$, is the linear subspace:*

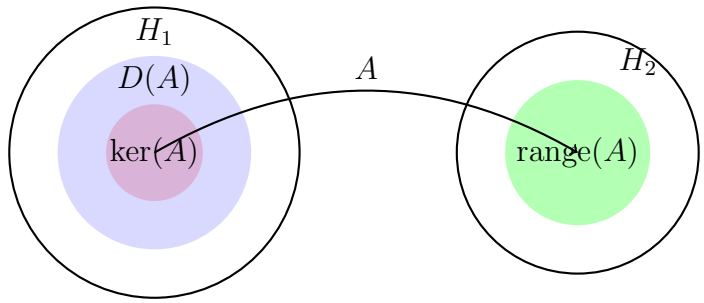
$$\ker(A) = \{x \in D(A) : A(x) = 0_{H_2}\}$$

where 0_{H_2} is the zero vector in H_2.

Definition 1.5 (Range of a Linear Operator)**.** *Let $A : D(A) \subset H_1 \to H_2$ be a linear operator from $D(A)$ to a vector space Y. The range, or image, of A, denoted by range(A) or Im(A), is the linear subspace:*

$$range(A) = \{A(x) \in H_2 : x \in D(A)\}.$$

Visualization of the kernel and of the range of a linear operator.



Definition 1.6 (Product (composition) of unbounded Linear Operators)**.** *If*

$$A : D(A) \subset H_1 \to H_2$$

and

$$B : D(B) \subset H_2 \to H_3$$

are linear operators, then

$$D(BA) = \{x \in D(A) \mid (Ax) \in D(B)\}$$
$$(BA)(x) = B(Ax), \quad \forall x \in D(BA).$$

Definition 1.7 (Scalar product and norm on the domain $D(A)$)**.** *Let $A : D(A) \subset H_1 \to H_2$ be a linear closed operator from the Hilbert space H_1 to the space H_2.*

Let $(\cdot,\cdot)_j$ and $\|\cdot\|_j$, for $j = 1, 2$, be the scalar product and the norm in H_j. We define the scalar product on $D(A)$ as

$$(x, y)_A := (x, y)_1 + (Ax, Ay)_2, \qquad \forall x, y \in D(A)$$

which gives the norm

$$\|x\|_A = (\|x\|_1^2 + \|Ax\|_2^2)^{1/2}, \qquad \forall x \in D(A)$$

is called the graphp norm, that is equivalent to the norm

$$\|x\|'_A = \|x\|_1 + \|Ax\|_2 \quad \forall x \in D(A)$$

Sometimes this last norm is called the graph norm.

Definition 1.8 (The Graph)**.** *The graph of an operator $A : D(A) \subset H_1 \to H_2$ is defined as*

$$G(A) = \{(x, Ax) \;\; x \in D(A)\}.$$

Definition 1.9 (Definition of Closed operator)**.** *An operator*

$$A : D(A) \subset H_1 \to H_2$$

is called closed it its graph $G(A)$ is a closed subset in the Hilbert spaces $H_1 \times H_2$.

Definition 1.10 (Definition of Closable operator)**.** *An operator $A : D(A) \subset H_1 \to H_2$ is called closable (or pre-closed) if there exists a closed operator B*

$$B : D(B) \subset H_1 \to H_2$$

such that

$$A \subset B.$$

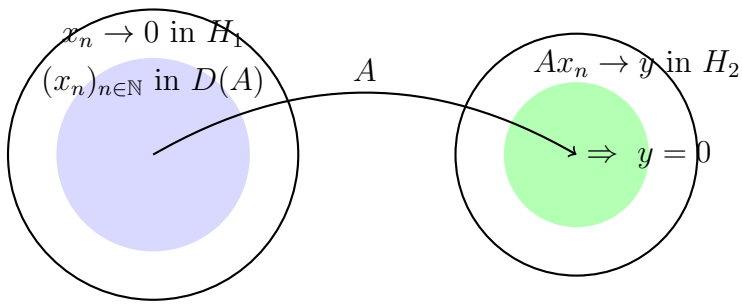
Theorem 1.11 (Characterization of closed operators)**.** *The following statements are equivalent*

(1) *A is closed (i.e. its graph $G(A)$ is a closed),*
(2) *For any sequence $(x_n)_{n\in\mathbb{N}}$ in $D(A)$ such that $x_n \to x$ in H_1 and $Ax_n \to y$ in H_2, it holds that $x \in D(A)$ and $Ax = y$,*
(3) *$(D(A), \|\cdot\|_A)$ is complete, or equivalently it is a Hilbert space.*

Theorem 1.12 (Characterization of a closable operator)**.** *The following statements are equivalent:*

(I) *A is closable (i.e. closable if there exists a closed operator $B : D(B) \subset H_1 \to H_2$ such that $A \subset B$),*
(II) *For any sequence $(x_n)_{n\in\mathbb{N}}$ in $D(A)$ such that $x_n \to 0$ in H_1 and $Ax_n \to y$ in H_2, it holds that $y = 0$,*
(III) *The closure of the graph $G(A) = \{(x, Ax) \quad x \in D(A)\}$ is the graph of a linear operator.*

Visualization of Closable operators



Visualization For any sequence $(x_n)_{n\in\mathbb{N}}$ in $D(A)$ such that $x_n \to 0$ in H_1 and $Ax_n \to y$ in H_2, it holds that $y = 0$.

Remark 1.13. *Since we have the characterization of closable operators see Theorem 1.12 from points (I) and (III): A is closable if and only if the closure of the graph*

$$G(A) = \{(x, Ax) \quad x \in D(A)\}$$

is the graph of a linear operator.

We have the following facts:

- *Suppose that A is closable*
- *and let $(\overline{A}, D(\overline{A}))$ be the closed linear operator such that*

$$\overline{G(A)} = G(\overline{A})$$

by the characterization of closable operator above.

Remark 1.14. *By the definition of $\overline{A}$ its domain $D(\overline{A})$ is defined as:*

- *$D(\overline{A})$ is the set of vectors x in H_1 such that there exists a sequence*

$$(x_n) \in D(A) \quad \text{such that} \quad x = \lim_n x_n \quad \text{in } H_1$$

and

$$(Ax_n) \ \ converges \ \ in \ \ H_2.$$

- *Operator $\overline{A}$ is defined as: For such sequence (x_n) we then set*

$$\overline{A}(x) := \lim_n A(x_n).$$

Since $\overline{G(A)}$ is the is the graph of an operator this definition of $\overline{A}(x)$ is independent of the sequence (x_n).

With respect to the inclusion of operators $A \subset \overline{A}$ is the smallest closed extension of a closable operator.

Definition 1.15 (Definition of closure of an operator)**.** *The operator $\overline{A}$ in the above remarks is called the closure of the closable operator A.*

2. The adjoint of an unbounded operator

Unlike bounded linear operators, where the concept of the adjoint operator is relatively straightforward, unbounded operators require careful consideration of the adjoint operator's domain. The explicit characterization of the domain of adjoint operators is notably complex and nontrivial.

Definition 2.1 (Definition of the adjoint operator: The domain)**.** *Let $(H_1, (\cdot,\cdot)_1$ a nd $(H_2, (\cdot,\cdot)_2$ be Hilbert spaces. Let $A : D(A) \subset H_1 \to H_2$ be a linear operator with dense domain $D(A)$ in H_1. The domain of the adjoint operator $D(A^*)$ is define as:*

$$D(A^*) := \{y \in H_2 \text{ such that there exists}$$

$$u \in H_1 \text{ such that} \quad (Ax, y)_2 = (x, u)_1 \quad \forall x \in D(A)\}.$$

Remark 2.2. *By the Riesz' representation theorem a vector $y \in H_2$ belongs to $D(A^*)$ if and only if the map*

$$x \mapsto (Ax, y)_2$$

is a linear and continuous functional on $(D(A), \|\cdot\|_1)$ or equivalently if there exists $c_y > 0$ such that

$$|(Ax, y)_2| \leq c_y \|x\|_1 \quad \forall x \in D(A).$$

Definition 2.3 (Definition of the adjoint operator: the operator A^*.)**.** *Since we have assumed that the domain $D(A)$ is dense in H_1 the vector $u \in H_1$ that satisfies*

$$(Ax, y)_2 = (x, u)_1 \quad \forall x \in D(A)$$

is uniquely determined by y. So setting

$$A^* y = u$$

we get a well defined linear map

$$A^* : D(A^*) \subset H_2 \to H_1.$$

Definition 2.4 (Definition of the adjoint operators)**.** *The operator* $(A^*, D(A^*))$ *is called the adjoint of* $(A, D(A))$*, that is we have*

$$(Ax, y)_2 = (x, A^*y)_1 \quad \forall x \in D(A),\ y \in D(A^*).$$

Remark 2.5. *The explicit description of* $D(A^*)$ *is in general a very difficult problem.*

Remark 2.6 (Visualization of the adjoint of $A : D(A) \subseteq H_1 \to H_2$)**.** *Let*

$$A : D(A) \subset H_1 \to H_2,$$

and

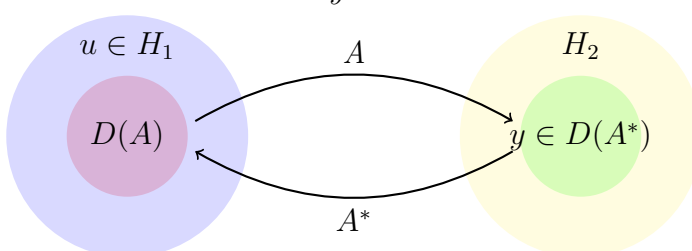
$$A^* : D(A^*) \subset H_2 \to H_1,$$

with

$$D(A^*) := \{y \in H_2 \ s.t.\ \exists\ u \in H_1 \ \ s.t.\ (Ax, y)_2 = (x, u)_1\ \forall x \in D(A)\}$$

then

$$A^*y := u$$



Definition 2.7 (Classes of operators)**.** *Let* $A : D(A) \subset H \to H$ *be a densely defined operator on* H*. We say that*

- A *is symmetric if* $A \subseteq A^*$*.*
- A *is self-adjoint if* $A = A^*$ *(this means that also* $D(A) = D(A^*)$*).*
- A *is essentially self-adjoint if its closure* $\overline{A}$ *is self-adjoint.*

Remark 2.8. *The domain* $D(A^*)$ *may not be dense in* H_2*.*

3. On overview on density theorems

Remark 3.1. *One can easily recognize if the domain* $D(A)$ *of an operator is dense in* L^p*, by the density theorems of functions spaces*

Theorem 3.2. *Step functions and continuous functions are dense in* $L^p[0, 1]$*,* $1 \leq p < \infty$*.*

Definition 3.3. *Let* $C_0(\Omega)$ *consists of all those continuous functions in which have compact support in* $\Omega \subset \mathbb{R}^n$ *with smooth boundary.*

Theorem 3.4. *Then the space* $C_0(\Omega)$ *is dense in* $L^p(\Omega)$ *for* $1 \leq < \infty$*.*

Definition 3.5. *Let* $C_0^\infty(\Omega)$ *consists of all those* $C^\infty(\Omega)$ *functions which have compact support in* $\Omega \subset \mathbb{R}^n$ *with smooth boundary.*

Theorem 3.6. *The space* $C_0^\infty(\Omega)$ *is dense in* $L^p(\Omega)$ *for* $1 \leq p < \infty$*.*

Definition 3.7. *The Schwartz class $S(\mathbb{R}^n)$ is defined by*

$$S(\mathbb{R}^n) = \{f \in C^\infty(\mathbb{R}^n) \text{ such that}$$
$$\sup_{x\in\mathbb{R}^n} |x^\alpha D^\beta f(x)| < \infty \text{ for any multi} - \text{indices } \alpha \textit{ and } \beta\}.$$

Roughly speaking, the Schwartz class consists of smooth functions whose derivatives decrease to zero faster than the inverse of any polynomial.

Theorem 3.8. *For any $1 \leq p < \infty$, the Schwartz class $S(\mathbb{R}^n)$ is a dense subspace of $L^p(\mathbb{R}^n)$.*

4. Example: The multiplication operators by continuous functions

- Let $I = [a, b] \subseteq \mathbb{R}$ be an interval.
- Given a continuous function $\varphi : I \to \mathbb{C}$ we define the multiplication operator M_φ on the Hilbert space $L^2(I)$ by

$$(M_\varphi)(x) := \varphi(x)f(x)$$
$$D(M_\varphi) = \{f \in L^2(I) \ \mid \ \varphi f \in L^2(I).\}$$

Theorem 4.1. *$D(M_\varphi)$ is dense in L^2 so the adjoint exists.*

Proof. Since $D(M_\varphi)$ contains $C_0(I)$ (continuous functions with compact support) and $C_0(I)$ is dense in L^2 □

Remark 4.2. *Since $D(M_\varphi)$ is dense in L^2 we have that the adjoint exists and is well defined.*

Theorem 4.3. *We have $(M_\varphi)^* = (M_{\overline{\varphi}})$.*

Proof. Observe that

$$\begin{aligned} D(M_\varphi) &= \{f \in L^2(I) \ \mid \ \varphi f \in L^2(I)\} \\ &= \{f \in L^2(I) \ \mid \ \overline{\varphi} f \in L^2(I)\} = D(M_{\overline{\varphi}}). \end{aligned}$$

For every f, $g \in D(M_\varphi) = D(M_{\overline{\varphi}})$ from the relation

$$(M_\varphi f, g) = \int_I \varphi(x)f(x)\overline{g(x)}dx = \int_I f(x)\overline{\overline{\varphi(x)}g(x)}dx = (f, M_{\overline{\varphi}}g)$$

We conclude that

$$M_{\overline{\varphi}} = (M_\varphi)^* \ \text{ on } \ D(M_\varphi)$$

that is

$$M_{\overline{\varphi}} \subseteq (M_\varphi)^*.$$

We have to prove the other inclusion, i.e.,

$$M_{\overline{\varphi}} \supseteq (M_\varphi)^*$$

so we take $g \in D((M_\varphi)^*)$ and set $h = (M_\varphi)^* g$. Let χ_K the characteristic function of the compact sets in I.

For $f \in D(M_\varphi)$ it is $\chi_K f \in D(M_\varphi)$ and

$$(M_\varphi(\chi_K f), g) = (\varphi \chi_K f, g) = (\chi_K f, h)$$

so that

$$\int_I \chi_K f(\varphi \overline{g} - \overline{h}) dx = 0$$

Since $D(M_\varphi)$ is dense in L^2 the element $\chi_K(\varphi \overline{g} - \overline{h})$ of L^2 must be zero so that

$$\varphi \overline{g} = \overline{h} \quad \Rightarrow \quad \overline{\varphi} g = h \quad on \quad K$$

and so on the entire I. That is we have $g \in D(M_{\overline{\varphi}})$ and

$$h = (M_\varphi)^* g = (M_{\overline{\varphi}}) g.$$

This proves that

$$M_{\overline{\varphi}} \supseteq (M_\varphi)^*.$$

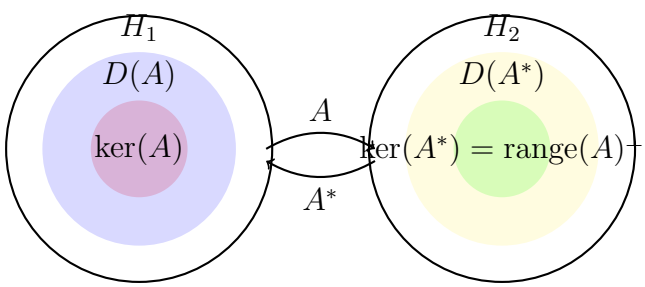
□

Remark 4.4. *When $\varphi(x) = x$ for $I = \mathbb{R}$ we have the position operator in Quantum Mechanics. This operator turns out to be self-adjoint on $L^2(\mathbb{R})$.*

5. Properties of the adjoint operator

Theorem 5.1 (The Adjoint)**.** *Let $A : D(A) \subset H_1 \to H_2$ and $B : D(B) \subset H_1 \to H_2$ such that $D(A)$ is dense in H_1. Then we have the following statements:*

- *A^* is a closed linear operator from H_2 to H_1.*
- *$range(A)^\perp = \ker(A^*)$.*
- *If $D(A^*)$ is dense in H_2, then $A \subseteq A^{**}$ where $A^{**} := (A^*)^*$.*
- *If $A \subseteq B$, then $A^* \supseteq B^*$.*



Visualization: $\text{range}(A)^\perp = \ker(A^*)$.

Theorem 5.2 (Adjoint and sum of operators)**.** *Let*

$$A : D(A) \subset H_1 \to H_2$$

and

$$B : D(B) \subset H_1 \to H_2$$

such that $D(A)$ is dense in H_1. Then we have the following statements:

- *$(\lambda A)^* = \overline{\lambda} A^*$ for all $\lambda \in \mathbb{C}$.*

- *If $D(A+B)$ is dense in H_1, then $(A+B)^* \supseteq A^* + B^*$.*
- *If B is a bounded operator with $D(B) = H_1$, then $(A+B)^* = A^* + B^*$.*

Theorem 5.3 (Adjoint and product of operators)**.** *Let*

$$A : D(A) \subset H_1 \to H_2$$

and

$$B : D(B) \subset H_2 \to H_3$$

such that $D(BA)$ is dense in H_1. Then we have the following statements:

- *If $D(B)$ is dense in H_2, then*
$$(BA)^* \supseteq A^* B^*$$
- *If B is a bounded operator with $D(B) = H_2$, then*
$$(BA)^* = A^* B^*.$$

Theorem 5.4 (A closable, adjoint and inverse operators)**.** *Let*

$$A : D(A) \subset H_1 \to H_2$$

such that $D(A)$ is dense in H_1. Then we have the following statements:

- *A is closable if and only if $D(A^*)$ is dense in H_2*
- *If A is closable, then $(\overline{A})^* = A^*$, and setting $A^{**} := (A^*)^*$ it is*
$$\overline{A} = A^{**}$$
- *A is closed if and only if $A = A^{**}$,*
- *Assume that A is closable and $ker(A) = \{0\}$ and $range(A)$ is dense in H_2. Then A^* is invertible and*
$$(A^*)^{-1} = (A^{-1})^*$$
- *It A is invertible , then A is closed if and only if A^{-1} is closed.*

6. Sobolev spaces in terms of Absolutely Continuous Functions

Definition 6.1 (Absolutely Continuous Functions)**.** *We assume $a, b \in \mathbb{R}$ with $a < b$. A function f on $[a, b]$ is called absolutely continuous if for each $\varepsilon > 0$, there exists a $\delta > 0$ such that*

$$\sum_{k=1}^{n} |f(b_k) - f(a_k)| < \varepsilon$$

for every finite family of pairwise disjoint subintervals (a_k, b_k) of $[a, b]$ satisfying

$$\sum_{k=1}^{n} (b_k - a_k) < \delta.$$

Definition 6.2. *The set of functions that are absolutely continuous on $[a, b]$ is denoted by $AC[a, b]$.*

Theorem 6.3 (Absolutely Continuous Functions and bounded variation functions)**.** *If $f \in AC[a,b]$, then $f \in C^0[a,b]$ and $f \in BV[a,b]$.*

Theorem 6.4. *The function $f \in AC[a,b]$ is and only if there exists a function $h \in L^1(a,b)$ such that*

$$f(x) - f(a) = \int_a^x h(t)dt, \quad for \quad x \in [a,b].$$

In this case f is differentiable a.e. and we have $f'(x) = h(x)$ a.e. in $[a,b]$. The function $h \in L^1(a,b)$ is uniquely determined by f.

Theorem 6.5 (Integration by parts for Absolutely Continuous Functions)**.** *For every f and $g \in AC[a,b]$ the integration by parts formula holds*

$$\int_a^b f'(t)g(t)dt + \int_a^b f(t)g'(t)dt = f(b)g(b) - f(a)g(a)$$

Let (a,b) be a bounded interval. The Sobolev spaces $H^n(a,b)$ and $H_0^n(a,b)$ can be expressed in terms of $AC[a,b]$ functions.

Definition 6.6. *We consider just $n = 1$ and $n = 2$*

$$H^1(a,b) = \{f \in AC[a,b] \ \mid \ f' \in L^2(a,b)\}$$
$$H^2(a,b) = \{f \in C^1[a,b] \ \mid \ f' \in H^1(a,b)\}$$
$$H_0^1(a,b) = \{f \in H^1(a,b) \ \mid \ f(a) = f(b) = 0\}$$
$$H_0^2(a,b) = \{f \in H^2(a,b) \ \mid \ f(a) = f(b) = 0 \ \ and \ \ f'(a) = f'(b) = 0\}$$

7. The adjoints of $Af := -if'$ and $A^2 f = -f''$

Let (a,b) be a bounded interval and let us defined

$$Af := -if'$$

on the Hilbert space $L^2(a,b)$ with domain

$$D(A) := H_0^1(a,b) = \{f \in H^1(a,b) \ \mid \ f(a) = f(b) = 0\}.$$

The square of A acts as

$$A^2 f := -f''$$

it domain is

$$D(A^2) = H_0^2(a,b) = \{f \in H^2(a,b) \ \mid \ f(a) = f(b) = 0 \ \ and \ \ f'(a) = f'(b) = 0\}$$

since

$$D(AB) = \{f \in D(B) \ \mid \ Bf \in D(A)\}$$

if $A = B$ we have

$$D(A^2) = \{f \in D(A) \ \mid \ Af \in D(A)\} = H_0^2(a,b).$$

Lemma 7.1. *The domains $D(A)$ and $D(A^2)$ are dense in $L^2(a,b)$.*

Proof. Observe that $D(A)$ and $D(A^2)$ contain continuous functions, which are dense in $L^2(a,b)$. $\square$

We describe the adjoint of these two operators.

Lemma 7.2. *Let $A : D(A) \subset L^2(a,b)$ where (a,b) is an interval and let us defined the operator*

- $Af := -if'$
- *with domain* $D(A) := H_0^1(a,b) = \{f \in H^1(a,b) \ \mid \ f(a) = f(b) = 0\}$.

Then, we have that

- $D(A^*) = H^1(a,b)$
- $A^*g := -ig'$.

Remark 7.3. *The proof is based on the integration by parts formulas: For every f and $g \in AC[a,b]$*

$$\int_a^b f'(t)g(t)dt + \int_a^b f(t)g'(t)dt = f(b)g(b) - f(a)g(a)$$

and on the fact that

$$(range(A))^\perp \text{ is contained in } \mathbb{C} \cdot 1.$$

Remark 7.4. *From the theorem it follows that $A \subset A*$ since*

$$D(A) := H_0^1(a,b) \subset H^1(a,b) = D(A^*)$$

and the formal operator is the same $-iD_x$. This means that the operator A is closable.

Remark 7.5. *On can prove that $A = A^{**}$. This means that A is closed. (since A is closed if and only if $A = A^{**}$)*

Lemma 7.6. *Let $A : D(A) \subset L^2(a,b)$ where (a,b) is an interval and let us defined the operator*

- $Af := -f''$
- *with domain* $D(A^2)$

given by

$$D(A^2) := H_0^2(a,b) = \{f \in H^2(a,b) \ \mid \ f(a) = f(b) = f'(a) = f'(b) = 0\}.$$

Then, we have that

- $D((A^2)^*) = H^2(a,b)$
- $(A^2)^*g := -g''$.

8. Criteria for Self-adjointness

Recall that A is symmetric if $A \subset A^*$ and A is self-adjoint if $A = A^*$.

Theorem 8.1. *Criteria of self-adjointness Let $(A, D(A))$ be a densely defined operator on a Hilbert spaces H. The following are equivalent:*

- $A = A^*$
- *A is closed and* $\ker(A^* + i) = \ker(A^* - i) = \{0\}$
- $\text{range}(A + i) = \ker(A - i) = H$

Corollary 8.2. *Let $(A, D(A))$ be asymmetric. The following are equivalent:*

- *A is essentially self-adjoint.*
- $\ker(A^* + i) = \ker(A^* - i) = \{0\}$
- $\overline{\text{range}(A + i)} = \overline{\ker(A - i)} = H$

Corollary 8.3. *Let $(A, D(A))$ be asymmetric. Then A is self-adjoint if and only if $\sigma(A)$ is real and in this case*

$$\|(A - \lambda)^{-1}\| \leq (Im(\lambda)^{-1}.$$

Definition 8.4 (B bounded relative to A)**.** *Let $(A, D(A))$ and $(B, D(B))$ be densely defined. We say that B bounded relative to A iff $D(A) \subset D(B)$ and there exist constants a, $b \geq 0$ such that*

$$\|Bx\| \leq a\|Ax\| + b\|x\|, \qquad \forall x \in D(A)$$

Theorem 8.5 (Kato-Rellich)**.** *Let $(A, D(A))$ be self-adjoint and $(B, D(B))$ symmetric. If B is bounded relative to A with bound $a < 1$, then $A + B$ is self-adjoint on $D(A)$.*

9. Spectral Theorem for Unbounded operators as functional calculus (version 1)

Theorem 9.1 (Theorem for unbounded self-adjoint operator)**.** *Let $A : D(A) \subset H \to H$ be an unbounded self-adjoint operator, where H is a complex separable Hilbert space H. Then there exists a unique function $\psi : B(\mathbb{R}) \to B(H)$ that has the following properties:*

(I) $\|\psi(g)\|_{B(H)} \leq \|g\|_\infty, \quad \forall g \in B(\mathbb{R})$.

(II) *ψ is a $*$-homomorphism.*

(III) *If g_n and $g \in B(\mathbb{R})$ with $g_n \to g$ for $t \in \mathbb{R}$ and $\sup_{n\in\mathbb{N}} |g_n| < 0$, then*

$$\psi(g_n)x \to \psi(g)x, \quad \forall x \in H.$$

Remark 9.2. *The proof of the spectral theorem is based in the Crucial Lemma in the following. We omit the proof of the spectral theorem for unbounded self-adjoint operators in the functional calculus version. The proof for interested readers is in Lecture 9.*

Lemma 9.3 (Crucial Lemma)**.** *Let $A : D(A) \subset H \to H$ be an unbounded self-adjoint operator, where H is a complex separable Hilbert space H. Then, there exist closed subspaces H_n, $n \in \mathbb{N}$ with the following properties:*

(A) *The subspaces are orthogonal to each other* $H_n \perp H_m$ *for* $n \neq m$ *and every vector* $x \in H$ *can be decomposed as*

$$x = \sum_{n=0}^{\infty} x_n \quad \textit{with} \quad x_n \in H_n,$$

(B) $H_n \subset D(A)$ *and* $A(H_n) \subset H_n$,

(C) *the restriction* A_n *of* A *to the subspace* H_n *is a linear bounded and self-adjoint operator.*

Remark 9.4. *We postpone the proof, it is quite technical.*